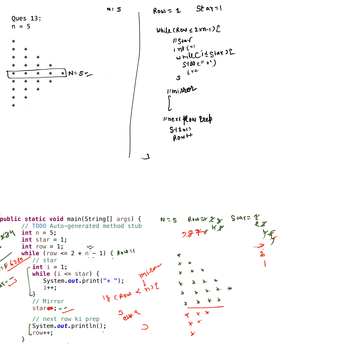

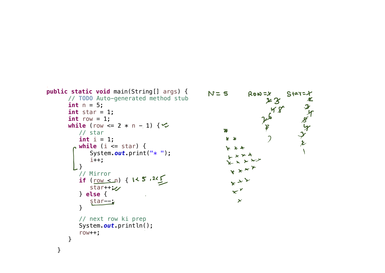

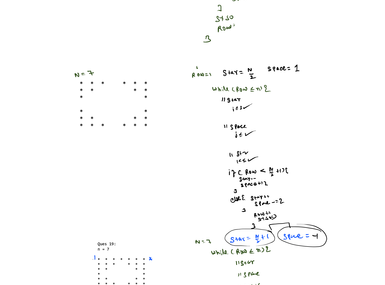

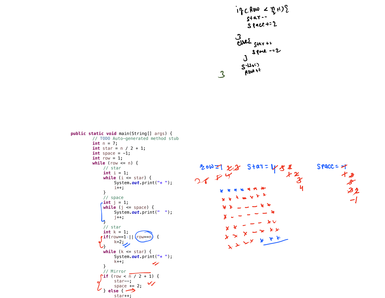

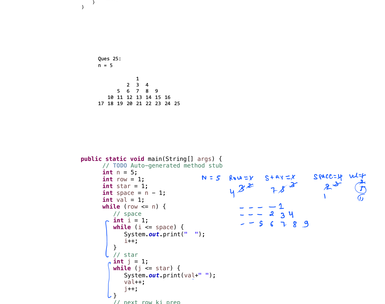

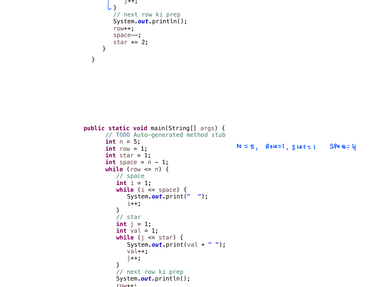

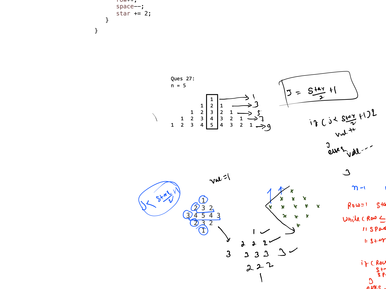

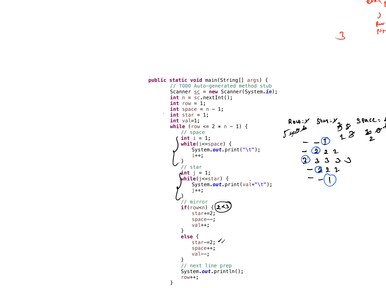

Lec-3

16/08/26 4:51 PM

Ques 13:
n = 5

```
*
*   *
*   *   *
*   *   *   *
*   *   *   *   *
*   *   *   *
*   *   *
*   *
*
```

N=5

N = 5

Row = 1 Star=1

while (Row ≤ 2×n-1) {
//star
int i=1
while(i ≤ star) {
Sysout("*")
i++
}
//mirror
{
//next Row Prep
Sysout()
Row++
}

```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int n = 5;
	int star = 1;
	int row = 1;
	while (row <= 2 * n - 1) {
		// star
		int i = 1;
		while (i <= star) {
			System.out.print("* ");
			i++;
		}
		// Mirror
		star++;
		// next row ki prep
		System.out.println();
		row++;
	}
}
```

N=5 Row=~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 6 7 8 9 Star=~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 4 3 2 1

Row=1, if (Row < n) {, mirror, 3 else { }

```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int n = 5;
	int star = 1;
	int row = 1;
	while (row <= 2 * n - 1) {
		// star
		int i = 1;
		while (i <= star) {
			System.out.print("* ");
			i++;
		}
		// Mirror
		if (row < n) {
			star++;
		} else {
			star--;
		}

		// next row ki prep
		System.out.println();
		row++;
	}
}
```

1<5, 2<5

N=5 Row=~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ ~~6~~ ~~7~~ 8 9 Star=~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ ~~4~~ ~~3~~ 2 1

Ques 15:
n = 5

```
*  *  *  *  *
      *  *  *  *
            *  *  *
                  *  *
                        *
                  *  *
            *  *  *
      *  *  *  *
*  *  *  *  *
```

Row=1 Star=N Space=0

While (Row ≤ 2n-1) {
//Space
//Star
//mirror
if (Row < n) {
Star --
Space +=2
}
else {
Star ++
Space -=2
}
Stso
Row++
}

N=7

```
*  *  *     *  *  *
*  *           *  *
*                 *

*                 *
*  *           *  *
*  *  *     *  *  *
```

Row=1 Star = N/2 Space = 1

While (Row ≤ n) {
//Star
i≤s ✓
// Space
j≤ ✓
// Star
k≤s ✓
if (Row < N/2 +1) {
Star--
Space+=2
}
else { Star++
Space -=2
}
Row++
Sysoc()
}

N=7 Star = N/2 +1 Space = -1

While (Row ≤ n) {
//Star
//Space
//Star
if (Row < n/2 +1) {
Star--
Space+=2
}
else { Star++
Space -=2
}
Sysoc()
Row++
}

Ques 19:
n = 7

```
*  *  *  *  *  *  *
*  *  *     *  *  *
*  *           *  *
*                 *
*  *           *  *
*  *  *     *  *  *
*  *  *  *  *  *  *
```

```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int n = 7;
	int star = n / 2 + 1;
	int space = -1;
	int row = 1;
	while (row <= n) {
		// star
		int i = 1;
		while (i <= star) {
			System.out.print("* ");
			i++;
		}
		// space
		int j = 1;
		while (j <= space) {
			System.out.print("  ");
			j++;
		}
		// star
		int k = 1;
		if(row==1 || row==n) {
			k=2;
		}
		while (k <= star) {
			System.out.print("* ");
			k++;
		}
		// Mirror
		if (row < n / 2 + 1) {
			star--;
			space += 2;
		} else {
			star++;
			space -= 2;
		}
		// next row prep
		System.out.println();
		row++;
	}
}
```

Row=~~1~~ ~~2~~ ~~3~~ 2 4 5 4 Star=~~4~~ ~~3~~ ~~2~~ 1 2 3 4 Space=~~-1~~ 1 3 5 3 1 -1

Ques 25:
n = 5

```
            1
         2  3  4
      5  6  7  8  9
  10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25
```

```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int n = 5;
	int row = 1;
	int star = 1;
	int space = n - 1;
	int val = 1;
	while (row <= n) {
		// space
		int i = 1;
		while (i <= space) {
			System.out.print("  ");
			i++;
		}
		// star
		int j = 1;
		while (j <= star) {
			System.out.print(val+" ");
			val++;
			j++;
		}
		// next row ki prep
		System.out.println();
		row++;
		space--;
		star += 2;
	}

}
```

N=5 Row=~~1~~ ~~2~~ ~~3~~ 4 Star=~~1~~ ~~3~~ 7 8 Space=4 ~~3~~ 2 1 Val=~~1~~ 2 5

- - - - 1
- - - 2 3 4
- - 5 6 7 8 9

```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	int n = 5;
	int row = 1;
	int star = 1;
	int space = n - 1;
	while (row <= n) {
		// space
		int i = 1;
		while (i <= space) {
			System.out.print("  ");
			i++;
		}
		// star
		int j = 1;
		int val = 1;
		while (j <= star) {
			System.out.print(val + " ");
			val++;
			j++;
		}
		// next row ki prep
		System.out.println();
		row++;
		space--;
		star += 2;
	}

}
```

N=5, Row=1, Star=1 Space=4

Ques 27:
n = 5

```
            1
         1  2  1
      1  2  3  2  1
   1  2  3  4  3  2  1
1  2  3  4  5  4  3  2  1
```

1 → 1, 3, 5, 7, 9

J = Star/2 +1

if (j < Star/2 +1) {
val ++
}
else {
vall--
}

j < Star/2 +1

val=1

```
    1
  2 3 2
3 4 5 4 3
  2 3 2
    1
```

1
2 2 2
3 3 3 3 3
2 2 2
1

n-1 1

Row=1 Star=1 Space=n-1

While (Row ≤ 2×n-1) {
//Space
//Star
if (Row < n) {
Star+2
Space--
}
else { Star-=2
Space++
}
Row++
Star
}

```java
public static void main(String[] args) {
	// TODO Auto-generated method stub
	Scanner sc = new Scanner(System.in);
	int n = sc.nextInt();
	int row = 1;
	int space = n - 1;
	int star = 1;
	int val=1;
	while (row <= 2 * n - 1) {
		// space
		int i = 1;
		while(i<=space) {
			System.out.print("\t");
			i++;
		}
		// star
		int j = 1;
		while(j<=star) {
			System.out.print(val+"\t");
			j++;
		}
		// mirror
		if(row<n) {
			star+=2;
			space--;
			val++;
		}
		else {
			star-=2;
			space++;
			val--;
		}
		// next line prep
		System.out.println();
		row++;
	}
```

2<3

Row=~~1~~ ~~2~~ 5 4 3 Star=~~1~~ ~~3~~ 5 1 3 Space=~~2~~ 1 0 1 2 Val=~~1~~ 2 3 2 1

- - 1
- 2 2 2
3 3 3 3 3
- 2 2 2
- - 1